Shubham Innani

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SUMMARY

AI scientist with 6+ years of experience developing and translating AI tools for clinical applications, with expertise in deep learning, foundation models, multimodal data, vision-language models, and reproducible HPC-based workflows. Hands-on experience developing AI models across pathology, radiology, retinal imaging, dermatology, and cardiac cine imaging. Experience advancing AI from model development and multi-institutional validation to prospective evaluation in hospital workflows, including intellectual property activities and planning. Focused on bridging AI development with product strategy, clinical workflows, real-world evidence, implementation and adoption, and healthcare informatics to translate emerging technologies into scalable solutions with measurable impact on human health.

EDUCATION

Bachelor of Technology (B.Tech), Electronics & Telecommunication Engineering 2016 – 2020
Shri Guru Gobind Singhji Institute of Engineering and Technology; US equivalent: B.S. Nanded, MH, India

EXPERIENCE

Research Data Analyst Dec 2023 – Present
Indiana University Indianapolis, IN, USA

- Develop and validate AI models using large-scale, multi-institutional datasets; trained WHO-aligned diagnostic and prognostic models on 2,500+ cases from $>$25 clinical sites.
- Build and maintain reproducible HPC workflows for clinical translation; delivered a tool, now in prospective validation.
- Collaborate with clinicians, investigators, scientists, and external partners to define research questions, design analytical approaches, analyze results, and communicate findings in conferences, publications, and research reports.
- Contribute scientific ideas and draft technical sections for NIH and other externally funded research proposals, including aims framing, methodology, and preliminary results.

Research Specialist B / **CIS Visiting Associate** Aug 2022 – Nov 2023
University of Pennsylvania Philadelphia, PA

- Developed multimodal AI approaches integrating histopathology imaging and clinical data for prognostic modeling of glioblastoma, reaching AUC $\approx$ 0.75 across 188 patients from multi-institutional cohorts.
- Led data curation, methodology design, software implementation, validation, and visualization across the project; delivered reproducible code and analyses used for downstream validation.
- Collaborated with multidisciplinary teams of pathologists, clinicians, and researchers to design experiments and interpret outputs.

Assistant System Engineer Jan 2021 – Sep 2022
Tata Consultancy Services Pune, India

- Led QA for cloud migration of transaction modules (legacy to AWS, Java/Spring Boot) for a leading U.S. financial client; owned API and end-to-end transaction verification, ensuring parity between legacy and cloud systems.
- Led a 6-person testing component, improving deployment reliability and reducing post-migration incidents.
- Collaborated with cross-functional teams and business Subject Matter Experts to validate complex transaction flows and support compliance requirements during migration.

Data Science Intern Mar 2020 – Jun 2020
Nymo.ai (AiCitta Intelligent Technology) Bangalore, India

- Developed computer vision and edge-AI inference pipelines (6–8 fps on-device) for Automatic Number Plate Recognition and mask-detection applications.
- Implemented object detection and OCR workflows for real-time inference on edge devices, evaluated model performance, and supported rapid prototype-to-deployment delivery for client applications.

AWARDS

2026 **2**nd Place (€300 Prize), Early Prediction of Cardiotoxicity, EchoRisk Challenge, MICCAI 2026

2026 3rd Place (€150 Prize), Left Ventricular Dysfunction Classification, EchoRisk Challenge, MICCAI 2026

2026 8th Place, LVEF Estimation, EchoRisk Challenge, MICCAI 2026

2025 Best Paper Award in Center for Computational Biology and Bioinformatics Retreat (CCBB), Indiana University

2025 **5th Place**, LLM-based report generation in pathology challenge (REG), MICCAI 2025

2020 **2nd Place (₹5000 Prize)**, semantic segmentation challenge, ICVGIP (IIT Jodhpur / IIIT Hyderabad / Intel)

2020 Top 10, retinal fundus glaucoma challenge (REFUGE2), MICCAI 2020

2020 Top 10, age-related macular degeneration (AMD) challenge, IEEE ISBI 2020

2019 **1st Place (₹10000 Prize)**, semantic segmentation challenge, NCVPRIPG (IIIT Hyderabad / Intel)

2019 Travel Grant for 7th National Conference on Computer Vision, Pattern Recognition, Image Processing, and Graphics (NCVPRIPG)

PUBLICATIONS

Journal

2026 **Shubham Innani**, Carla Pitarch-Abaigar, Marwan M. Majeed, Hannah Harmsen, W. Robert Bell, Dimitrios Makris, Spyridon Bakas, AI-based prognostic risk stratification of adult-type diffuse glioma patients from H&E-stained slides, Cancers. 2026; 18(20):3259.

2026 Ruining Deng, Tianyuan Yao, Yucheng Tang, et al., **Shubham Innani**, Ujjwal Baid, et al., KPIs 2024 challenge: Advancing glomerular segmentation from patch- to slide-level, *Medical Image Analysis*, 114:104234.

2026 **Shubham Innani**, MacLean Nasrallah, W. Robert Bell, Bhakti Baheti*, Spyridon Bakas*, AI-driven WHO 2021 classification of gliomas based only on H&E-stained slides, *Neuro-Oncology*, 28(1):282–296.

2025 **Shubham Innani**, MacLean Nasrallah, W. Robert Bell, Bhakti Baheti*, Spyridon Bakas*, Interpretable AI based determination of glioma IDH mutation status directly from histology slides, *Neuro-Oncology Advances*, 7(1):vdaf140.

2025 Bhakti Baheti, Sunny Rai, **Shubham Innani**, Garv Mehdiratta, W. Robert Bell, Sharath Chandra Guntuku, MacLean P. Nasrallah, Spyridon Bakas, Multimodal Explainable Artificial Intelligence for Prognostic Stratification of Patients With Glioblastoma, *Modern Pathology*, 38(9):100797.

2025 Katrina Collins*, **Shubham Innani***, Kingsley Ebare, Mohammed Saad, Stephanie E. Siegmund, Sean R. Williamson, Fiona Maclean, Andres Matoso, Ankur Sangoi, Michelle S. Hirsch, Dibson D. Gondim, Andres M. Acosta, Bhakti Baheti, Spyridon Bakas, Muhammad T. Idrees, AI-Based Classification of Renal Oncocytic Neoplasms: Advancing From a 2-Class to a 3-Class Model Including Chromophobe Renal Cell Carcinoma, *Archives of Pathology & Laboratory Medicine*, doi:10.5858/arpa.2024-0374-OA (*equal contribution).

2024 Bhakti Baheti, **Shubham Innani**, MacLean P. Nasrallah, Spyridon Bakas, Prognostic stratification of glioblastoma patients by unsupervised clustering of morphology patterns on whole slide images, *Frontiers in Neuroscience*, 18:1304191.

2023 **Shubham Innani**, Prasad Dutande, Ujjwal Baid, Venu Pokuri, Spyridon Bakas, Sanjay Talbar, Bhakti Baheti, Sharath Chandra Guntuku, Generative adversarial networks based skin lesion segmentation, *Scientific Reports*, 13(1):13467.

2022 Huihui Fang, Fei Li, et al., Ujjwal Baid, **Shubham Innani**, et al., ADAM Challenge: Detecting Age-Related Macular Degeneration From Fundus Images, *IEEE Transactions on Medical Imaging*, 41(10):2828–2847.

2020 Bhakti Baheti, **Shubham Innani**, Suhas Gajre, Sanjay Talbar, Semantic scene segmentation in unstructured environment with modified DeepLabV3+, *Pattern Recognition Letters*, 138:223–229.

Conference

2026 **Shubham Innani**, Siddhesh Thakur, Michael Feldman, Spyridon Bakas, Diagnostic Pathology Reporting and Reasoning via Autoregressive and Retrieval-Based Chain-of-Thought, in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2026 Workshops and Challenges*, Springer Nature Switzerland, LNCS 17252, 2026.

2026 **Shubham Innani**, Suhang You, Carla Pitarch-Abaigar, Dimitrios Makris, Spyridon Bakas, Rare-Class Conditional Routing with Foundation Model Representations for Glioma Histologic Sub-region Classification, in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2026 Workshops and Challenges*, Springer Nature Switzerland, LNCS 17252, 2026.

2026 **Shubham Innani**, Carla Pitarch-Abaigar, Spyridon Bakas, Scaling Native 3D Self-Supervised DINOv2 for Brain MRI Foundation Models, in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2026 Workshops and Challenges*, Springer Nature Switzerland, LNCS 17252, 2026.

2026 **Shubham Innani**, Michael Feldman, Spyridon Bakas, One Encoder to Address All: Cardio-Oncology Assessment of Left Ventricular Ejection Fraction, Left Ventricular Dysfunction, and Early Cardiotoxicity in Echocardiography Cine Loops, in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2026 Workshops and Challenges*, Springer Nature Switzerland, LNCS 17252, 2026.

2024 **Shubham Innani**, MacLean Nasrallah, W. Robert Bell, Bhakti Baheti[*], Spyridon Bakas[*], Multi-scale Whole Slide Image Assessment Improves Deep Learning based WHO 2021 Glioma Classification, *MICCAI Workshop on Computational Pathology with Multimodal Data (COMPAYL)*.

2024 **Shubham Innani**, Bhakti Baheti, MacLean Nasrallah, Spyridon Bakas, Weakly supervised IDH-status glioma classification from H&E-stained whole slide images, *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2022 **Shubham Innani**, Prasad Dutande, Bhakti Baheti, Ujjwal Baid, Sanjay Talbar, Deep Learning based Novel Cascaded Approach for Skin Lesion Analysis, *International Conference on Computer Vision & Image Processing (CVIP)*.

2021 Bhakti Baheti, **Shubham Innani**, Suhas Gajre, Sanjay Talbar, Pedestrian Detection and Movement Direction Recognition with CNN, *International Conference on Pattern Recognition and Machine Intelligence (PReMI)*.

2021 **Shubham Innani**, Prasad Dutande, Bhakti Baheti, Sanjay Talbar, Ujjwal Baid, Fuse-PN: A Novel Architecture for Anomaly Pattern Segmentation in Aerial Agricultural Images, *IEEE/CVF CVPR Workshops*, 2960–2968.

2020 Mang Tik Chiu, Xingqian Xu, Kai Wang, et al., Ujjwal Baid, **Shubham Innani**, Prasad Dutande, Bhakti Baheti, Sanjay Talbar, et al., The 1st Agriculture-Vision Challenge: Methods and Results, *IEEE/CVF CVPR Workshops*, 212–218.

2020 Bhakti Baheti[*], **Shubham Innani**[*], Suhas Gajre, Sanjay Talbar, Eff-UNet: A Novel Architecture for Semantic Segmentation in Unstructured Environment, *IEEE/CVF CVPR Workshops*, 1473–1481 ([*]equal contribution).

Abstract

2026 **Shubham Innani**, Alexander Toth, Marwan M. Majeed, Seemaab Ali, Hannah Harmsen, W. Robert Bell, Spyridon Bakas, PATH-62. Prospective evaluation of AI-based WHO 2021 adult-type diffuse glioma classification based solely on H&E-stained slides, *Neuro-Oncology*, 28(Suppl 5).

2025 **Shubham Innani**, Carla Pitarch-Abaigar, Marwan M. Majeed, Hannah Harmsen, W. Robert Bell, Dimitrios Makris, Spyridon Bakas, PATH-69. AI-based prognostic risk stratification of adult-type diffuse gliomas using H&E-stained slides, *Neuro-Oncology*, 27(Suppl 5):v257.

2025 **Shubham Innani**, W. Robert Bell, MacLean P. Nasrallah, Bhakti Baheti, Spyridon Bakas, PATH-67. AI-based WHO 2021 classification of adult-type diffuse glioma solely from H&E-stained slides, *Neuro-Oncology*, 27(Suppl 5):v256.

2025 Siddhesh Thakur, Sylwia Malec, Carla Pitarch-Abaigar, Akis Linardos, **Shubham Innani**, Sanyukta Adap, Shahriar Faghani, et al., IMG-121. BraTS-Pathology 2024: Insights and Future Directions Informed by the AI-RANO & RANO-RGP Effort to Assess Glioblastoma Heterogeneity, *Neuro-Oncology*, 27(Suppl 5):v303–v304.

2025 **Shubham Innani**, W. Robert Bell, MacLean Nasrallah, Bhakti Baheti, Spyridon Bakas, Abstract 6247. AI predicts 2021 WHO glioma subtypes from whole slide images, *Cancer Research*, 85(8 Suppl 1):6247.

2024 **Shubham Innani**, Bhakti Baheti, MacLean P. Nasrallah, W. Robert Bell, Spyridon Bakas, PATH-39. AI-based identification of glioma IDH mutational status from H&E-stained WSIs, *Neuro-Oncology*, 26(Suppl 8):viii187.

2024 Siddhesh Thakur, et al., **Shubham Innani**, Bhakti Baheti, et al., TMIC-60. BraTS-Path: assessing heterogeneous histopathologic regions in glioblastoma, *Neuro-Oncology*, 26(Suppl 8):viii312.

2023 **Shubham Innani**, Bhakti Baheti, MacLean P. Nasrallah, Spyridon Bakas, Interpretable IDH classification from H&E-stained histology slides, *Neuro-Oncology*, 25(Suppl 5):v177.

2023 Bhakti Baheti, Sunny Rai, **Shubham Innani**, Garv Mehdiratta, Sharath Chandra Guntuku, MacLean P. Nasrallah, Spyridon Bakas, Detecting Histologic & Clinical Glioblastoma Patterns of Prognostic Relevance, *Neuro-Oncology*, 25(Suppl 5):v126.

2023 Bhakti Baheti, **Shubham Innani**, Garv Mehdiratta, MacLean P. Nasrallah, Spyridon Bakas, Interpretable WSI

prognostic stratification of glioblastoma patients, *Neuro-Oncology*, 25(Suppl 2):ii103–ii104.

2023 Bhakti Baheti, **Shubham Innani**, MacLean P. Nasrallah, Spyridon Bakas, Unsupervised clustering of morphology patterns on WSIs guide prognostic stratification of glioblastoma patients, *Neuro-Oncology*, 25(Suppl 2):ii15.

Preprint

2026 **Shubham Innani**, Suhang You, Adam Shephard, Bhakti Baheti, Francesco Ciompi, et al., Explainable Artificial Intelligence (XAI) in Computational Pathology: Definitions, Taxonomy, and Recommendations, *arXiv*:2608.28820.

2026 Yumi Lee, Harim Oh, Hyoryung Kim, et al., **Shubham Innani**, et al., Benchmarking Vision-Language Models for Automated Pathology Diagnosis and Report Generation, *arXiv*:2609.00866.

2026 Sanket Kachole, Siddhesh Thakur, **Shubham Innani**, Sanyukta Adap, Suhang You, Carla Pitarch-Abaigar, Spyridon Bakas, Multi-FRuGaL: Multimodal Flexible Redundancy-aware Decomposed Gated Learning for Cancer Diagnosis and Prognosis, *arXiv*:2606.06867.

2024 Spyridon Bakas, Siddhesh P. Thakur, et al., **Shubham Innani**, Bhakti Baheti, et al., BraTS-Path Challenge: Assessing Heterogeneous Histopathologic Brain Tumor Sub-regions, *arXiv*:2405.10871.

2022 Huihui Fang, Fei Li, Huazhu Fu, et al., Ujjwal Baid, **Shubham Innani**, et al., REFUGE2 Challenge: Treasure for Multi-Domain Learning in Glaucoma Assessment, *arXiv*:2202.08994.

PRESENTATIONS

2026 Oral & Poster, 31th annual meeting of the Society for Neuro-Oncology (SNO 2026), Philadelphia, PA

2025 Oral & Poster, 30th annual meeting of the Society for Neuro-Oncology (SNO 2025), Honolulu, HI

2025 Oral presentation, 21st European Congress on Digital Pathology (ECDP 2025), Barcelona, Spain

2025 Poster, 116th annual meeting of the American Association for Cancer Research (AACR), Chicago, IL

2024 Oral, 29th annual meeting of the Society for Neuro-Oncology (SNO 2024), Houston, TX

2024 Two Oral presentations, 20th European Congress on Digital Pathology (ECDP 2024), Vilnius, Lithuania

2023 Oral & Poster, 19th European Congress on Digital Pathology (ECDP 2023), Budapest, Hungary

COMMUNITY ENGAGEMENTS

- **Reviewer:** IEEE Transactions on Medical Imaging, Scientific Reports, IEEE Signal Processing Letters, IEEE TNNLS, IEEE CVPR Workshops, IEEE ISBI, Elsevier Neurocomputing, MICCAI, npj Precision Oncology, BMC Cancer.
- **Program Committee:** 2nd International Workshop on Medical Optical Imaging and Virtual Microscopy Image Analysis (MOVI) @ MICCAI 2024.
- **Volunteer:** NCI Informatics Technology for Cancer Research (ITCR) annual meeting, 2024.
- **Member:** MICCAI SIG on Computational Pathology; Society for Neuro-Oncology (2025–2026); ECDP (2024–2026); AACR (2025); MICCAI Society (2024); IEEE Signal Processing Society (2024).

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